

popquiz adding vectors equilibrium

 This is a preview of the published version of the quiz

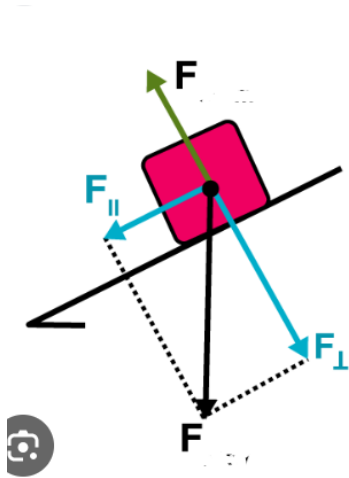
Started: Apr 29 at 7:57pm

Quiz Instructions

We discussed the problems in class, Can't help you



Question 1 14.3 pts



Here is a box sliding on an inclined plane. The x-axis is along the plane. the y-axis is perpendicular to the plane.

What is true?



the black vector is the y-component of the weight and the green vector is the normal force



the green vector is the y-component of the weight and the black vector is the normal



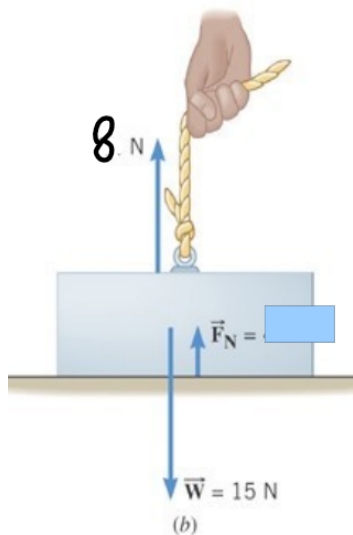
the green vector is the normal force and the black vector the weight



the black vector is the weight and there is no other force (green shows the direction of y-component)



Question 2 14.3 pts



What is the normal force. As discussed in class

☐

23N

☐

15N

☐

7N

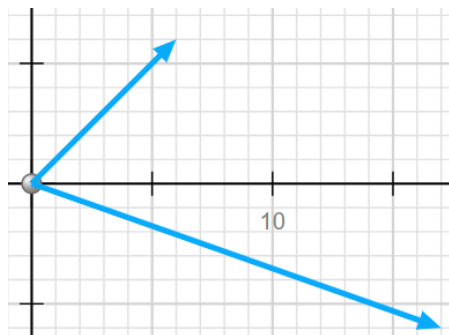
☐

0N

⋮

Question 3 14.3 pts

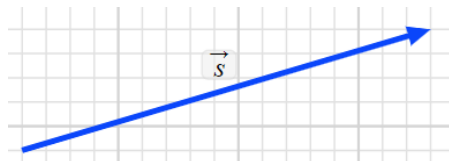
Here are 2 vectors A (1st quadrant) and B.

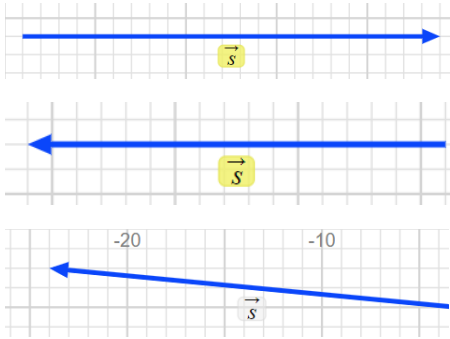


Use the parallelogram method or the tail to head method, to trace the vector sum S ($A + B$)

What is the vector sum? select one: (1,2,3,4). hint: make sure to get a parallelogram.

https://phet.colorado.edu/sims/html/vector-addition/latest/vector-addition_all.html




☐

1

☐

2

☐

4

☐

3

⋮

Question 4 14.3 pts

Building on the previous questions. A and B are 2 forces pulling a crate placed on a frictionless surface. If you want the crate to stay at rest, you need to introduce another force C (pull). So the crate is in static equilibrium. What is true.

☐

C is along the positive x-axis

☐

C is located in the first quadrant

☐

C is along the negative x-axis

☐

C is along the positive y-axis.

⋮

Question 5 14.3 pts

I class I showed you a video where I was doing a demo. I placed a weight on a scale (say weight is in grams. does not matter here). I pulled gently on the weight up.



what is true

☐

you are reading your pull

☐

you are reading the true weight

☐

you are reading the apparent weight / normal force

☐

I missed the class and didn't catch up

⋮

Question 6 14.3 pts

the normal force is the apparent weight

A brick (like skier) is resting on the surface of a flat board. As one end of the board is slowly raised, what change, if any, is there in the normal force exerted on the brick?

a) The normal force increases.

**What about the scale?
Its called the apparent weight**

b) The normal force decreases.

Same reading ?
Do demo

c) The normal force remains constant.

d) Only the direction of the normal force changes.



☐

a

☐

c

☐

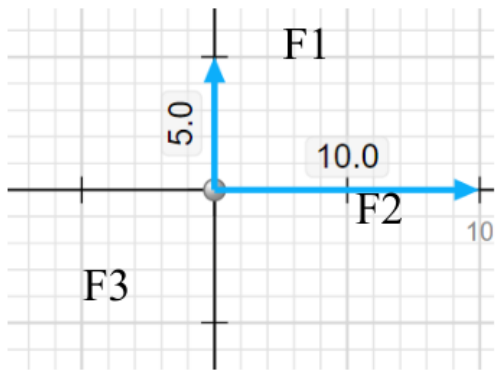
b

☐

d

☐

Question 7 14.3 pts



3 forces are acting on a ring placed on a force table. The ring is in equilibrium. We did it in class. what is the components of the third force so the ring stays at rest. Can't help you

☐

F3(15,15)

☐

F3(5,10)

☐

F3(-10,-5)

☐

No answer text provided.

☐

F3(10,5)

☐

can not be determined

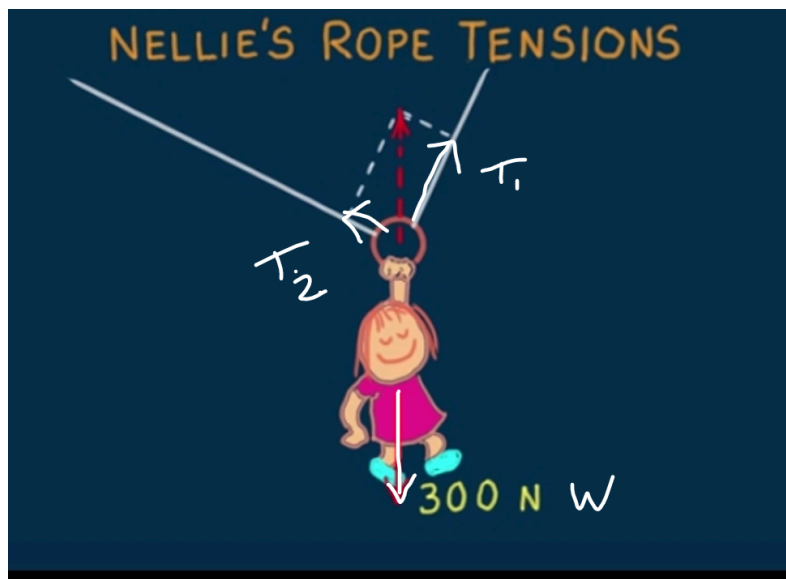
☐

I missed the class and didn't catch up

☐

Question 8 5 pts

bonus



Little Suzie is in equilibrium. Her weight is 300N. The tensions in the ropes are T_1 and T_2 . They are vectors. What is true

- ☐ magnitude of T_1 + magnitude of T_2 = 300
- ☐ magnitude of T_1 = magnitude of T_2 = 300
- ☐ y-component of T_1 + y-component of T_2 = 300
- ☐ x-component of T_1 + x-component of T_2 = 300

Not saved

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